

Batch: A2 Roll No.: 16010324044
Experiment No. 2
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date
Title : Analysis of the signal Using DFT
AIM : To find the frequency components of the signal using Discrete Fourier Transform, application of properties of DFT.

OUTCOME : Analyse discrete time systems using discrete Fourier transform

PROCEDURE:

The representation of a sequence by samples of its spectrum leads to Discrete Fourier Transform. DFT is powerful computational tool for performing frequency analysis of Discrete Time signal and linear filtering. Properties of DFTs are extremely helpful in the application of DFT to practical problems. FFT is computationally efficient algorithm for computing DFT and IDFT.

Write MATLAB program for

1. Find 4 point DFT of $x(n) = \{0,1,2,3\}$, Plot Magnitude and Phase Spectrum of the signal.

```

clc;
clear;
close all;

x = [0 1 2 3];

X = fft(x,4);

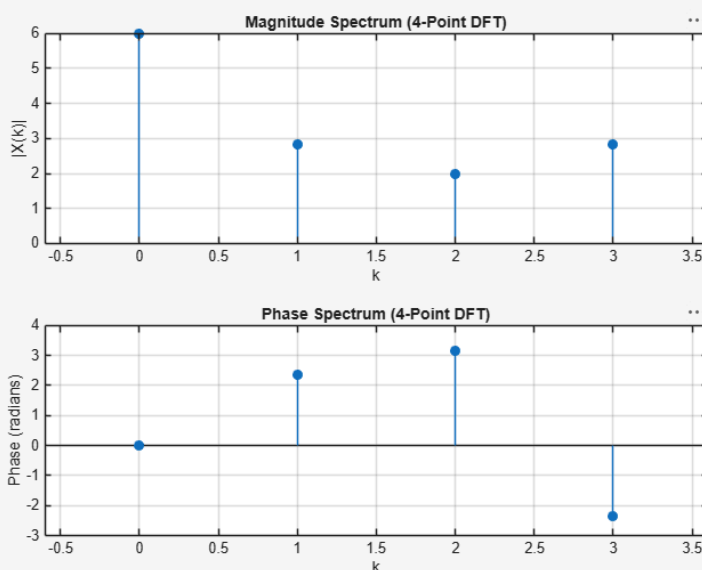
disp('4-Point DFT:');
disp(X);

figure;

subplot(2,1,1);
stem(0:3, abs(X), 'filled');
grid on;
title('Magnitude Spectrum (4-Point DFT)');
xlabel('k');
ylabel('|X(k)|');

subplot(2,1,2);
stem(0:3, angle(X), 'filled');
grid on;
title('Phase Spectrum (4-Point DFT)');
xlabel('k');
ylabel('Phase (radians)');

```



2. Find 2 point DFT of $x(n) = \{0,1,2,3\}$, Plot Magnitude and Phase Spectrum of the signal.

```

clc;
clear;
close all;

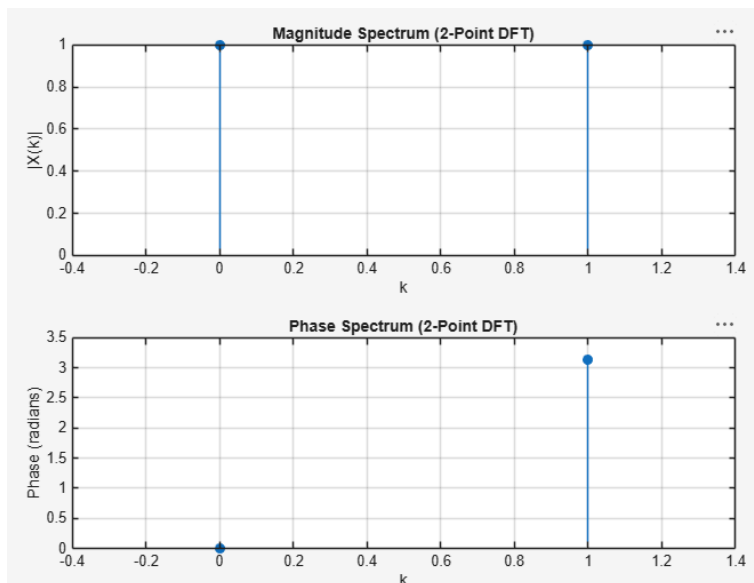
x = [0 1 2 3];
X = fft(x,2);

disp('2-Point DFT:');
disp(X);

figure;

subplot(2,1,1);
stem(0:1, abs(X), 'filled');
grid on;
title('Magnitude Spectrum (2-Point DFT)');
xlabel('k');
ylabel('|X(k)|');

subplot(2,1,2);
stem(0:1, angle(X), 'filled');
grid on;
title('Phase Spectrum (2-Point DFT)');
xlabel('k');
ylabel('Phase (radians)');
  
```



3. Find 8 point DFT of $x(n) = \{0,1,2,3\}$, Plot Magnitude and Phase Spectrum of the signal.

```

clc;
clear;
close all;

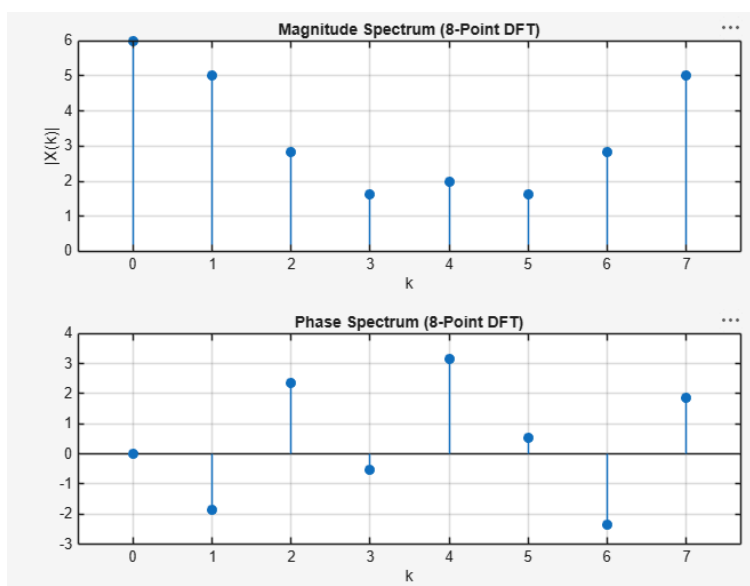
x = [0 1 2 3];
X = fft(x,8);

disp('8-Point DFT:');
disp(X);

figure;

subplot(2,1,1);
stem(0:7, abs(X), 'filled');
grid on;
title('Magnitude Spectrum (8-Point DFT)');
xlabel('k');
ylabel('|X(k)|');

subplot(2,1,2);
stem(0:7, angle(X), 'filled');
grid on;
title('Phase Spectrum (8-Point DFT)');
xlabel('k');
% ylabel('Phase (radians)');
  
```



4. Comment on the above three DFTs.

- **2-point DFT:** The sequence is truncated to the first two samples ($[0 \ 1]$) by MATLAB's `fft(x, 2)`, so the result differs from the theoretical aliasing approach often used in textbooks.
- **4-point DFT:** Produces the exact DFT of the original sequence.
- **8-point DFT:** Zero-pads the sequence to $[0 \ 1 \ 2 \ 3 \ 0 \ 0 \ 0 \ 0]$, resulting in a denser (higher-resolution) frequency spectrum.

5. Find DFT of $x(n) = \{0,0,0,0,1,2,3\}$, Plot Magnitude and Phase Spectrum of the signal.

```

clc;
clear;
close all;

x = [0 0 0 0 1 2 3];

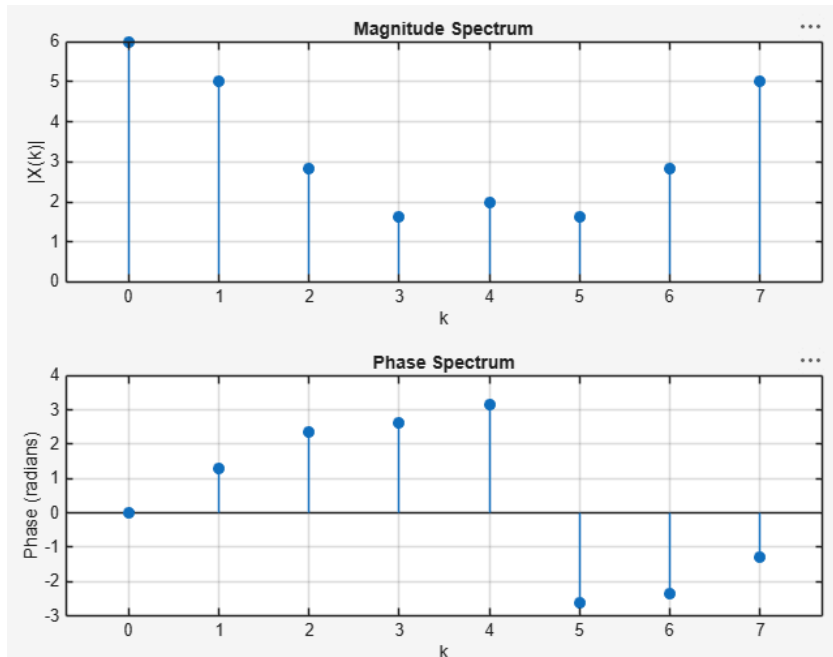
X = fft(x,8);

disp('DFT of x(n) = [0 0 0 0 1 2 3]');
disp(X);

figure;

subplot(2,1,1);
stem(0:7, abs(X),'filled');
grid on;
title('Magnitude Spectrum');
xlabel('k');
ylabel('|X(k)|');

subplot(2,1,2);
stem(0:7, angle(X),'filled');
grid on;
title('Phase Spectrum');
xlabel('k');
ylabel('Phase (radians)');
  
```



6. Find DFT of $x(n) = \{0,1,2,3,0,0,0,0\}$, Plot Magnitude and Phase Spectrum of the signal.

```

clc;
clear;
close all;

x = [0 1 2 3 0 0 0 0];

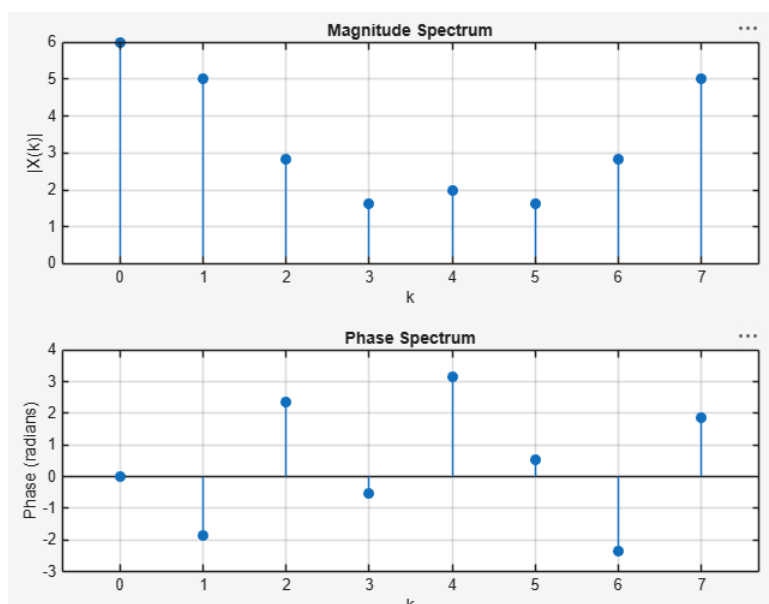
X = fft(x,8);

disp('DFT of x(n) = [0 1 2 3 0 0 0 0]');
disp(X);

figure;

subplot(2,1,1);
stem(0:7, abs(X),'filled');
grid on;
title('Magnitude Spectrum');
xlabel('k');
ylabel('|X(k)|');

subplot(2,1,2);
stem(0:7, angle(X),'filled');
grid on;
title('Phase Spectrum');
xlabel('k');
ylabel('Phase (radians)');
  
```



7. Find DFT of $x(n) = \{0,0,1,0,2,0,3,0\}$, Plot Magnitude and Phase Spectrum of the signal.

```

clc;
clear;
close all;

x = [0 0 1 0 2 0 3 0];

X = fft(x,8);

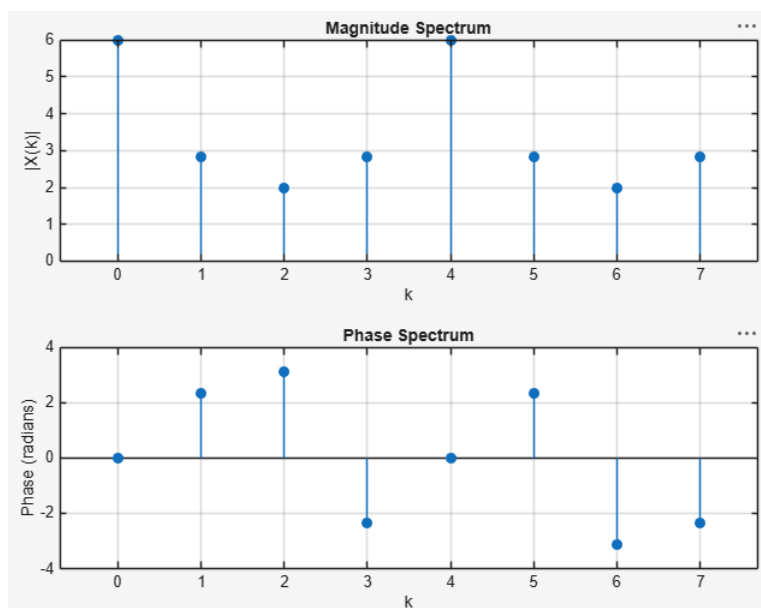
disp('DFT of x(n) = [0 0 1 0 2 0 3 0]');
disp(X);

figure;

subplot(2,1,1);
stem(0:7, abs(X), 'filled');
grid on;
title('Magnitude Spectrum');
xlabel('k');
ylabel('|X(k)|');

subplot(2,1,2);
stem(0:7, angle(X), 'filled');
grid on;
title('Phase Spectrum');
xlabel('k');
ylabel('Phase (radians)');

```



8. Find DFT of $x(n) = \{0,1,2,3,0,1,2,3\}$, Plot Magnitude and Phase Spectrum of the signal.

```

clc;
clear;
close all;

x = [0 1 2 3 0 1 2 3];

X = fft(x,8);

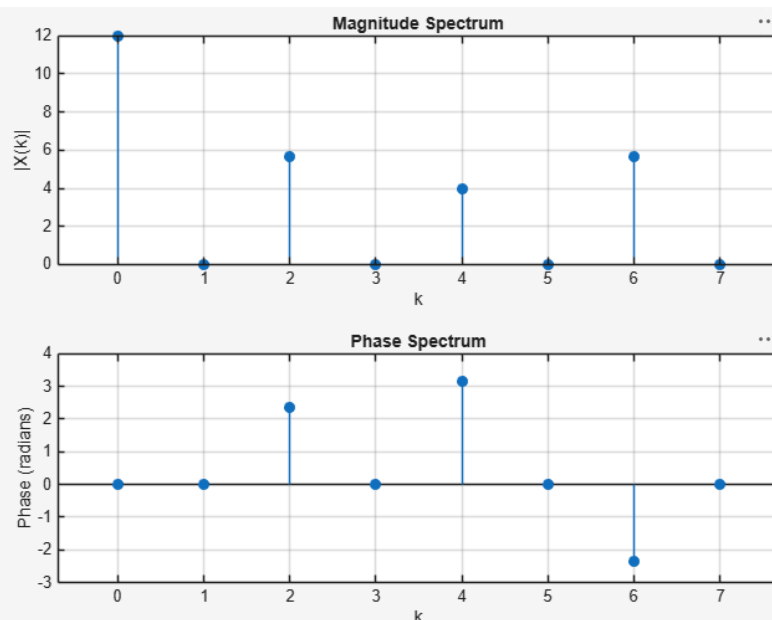
disp('DFT of x(n) = [0 1 2 3 0 1 2 3]');
disp(X);

figure;

subplot(2,1,1);
stem(0:7, abs(X), 'filled');
grid on;
title('Magnitude Spectrum');
xlabel('k');
ylabel('|X(k)|');

subplot(2,1,2);
stem(0:7, angle(X), 'filled');
grid on;
title('Phase Spectrum');
xlabel('k');
ylabel('Phase (radians)');

```



9. Comment on the DFTs of 5,6,7 and 8.

- **Q5 DFT:** The sequence is a **5-sample delayed version** of $[0\ 1\ 2\ 3\ 0\ 0\ 0\ 0][0\ 1\ 2\ 3\ 0\ 0\ 0\ 0][0\ 1\ 2\ 3\ 0\ 0\ 0\ 0]$. The magnitude spectrum remains the same as Q6, while the phase spectrum changes due to the time shift.
- **Q6 DFT:** The sequence **zero-pads** the original 4-point signal to $[0\ 1\ 2\ 3\ 0\ 0\ 0\ 0][0\ 1\ 2\ 3\ 0\ 0\ 0\ 0][0\ 1\ 2\ 3\ 0\ 0\ 0\ 0]$. Zero padding increases the number of frequency samples, producing a denser (higher-resolution) spectrum without changing the actual frequency content.
- **Q7 DFT:** Zeros are inserted between every sample, forming $[0\ 0\ 1\ 0\ 2\ 0\ 3\ 0][0\ 0\ 1\ 0\ 2\ 0\ 3\ 0][0\ 0\ 1\ 0\ 2\ 0\ 3\ 0]$. This corresponds to **upsampling by a factor of 2**, causing spectral replication in the frequency domain.
- **Q8 DFT:** The sequence repeats the original signal, forming $[0\ 1\ 2\ 3\ 0\ 1\ 2\ 3][0\ 1\ 2\ 3\ 0\ 1\ 2\ 3][0\ 1\ 2\ 3\ 0\ 1\ 2\ 3]$. Repeating the sequence doubles the signal length and produces non-zero DFT values mainly at even frequency bins, with magnitudes increased compared to the original sequence.

10. Find DFT of $x(n) = \{3,0,1,2\}$, Plot Magnitude and Phase Spectrum of the signal.

```
clc;
clear;
close all;

x = [3 0 1 2];

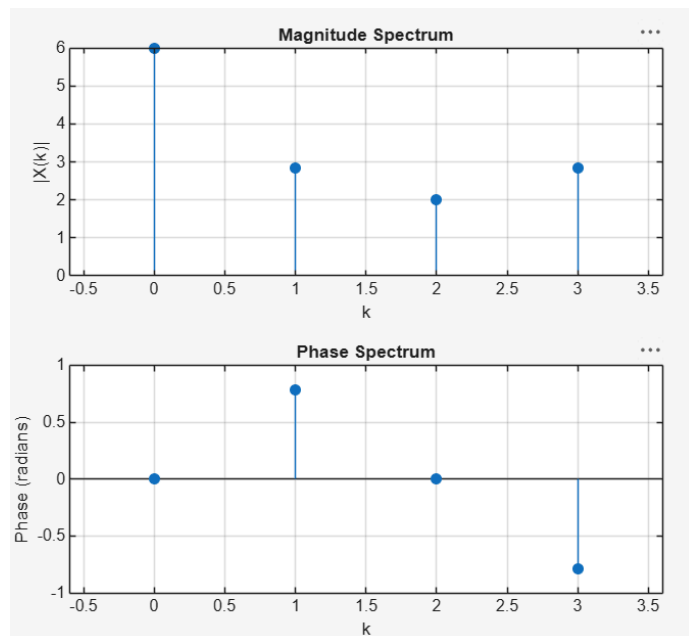
X = fft(x,4);

disp('DFT of x(n) = [3 0 1 2]');
disp(X);

figure;

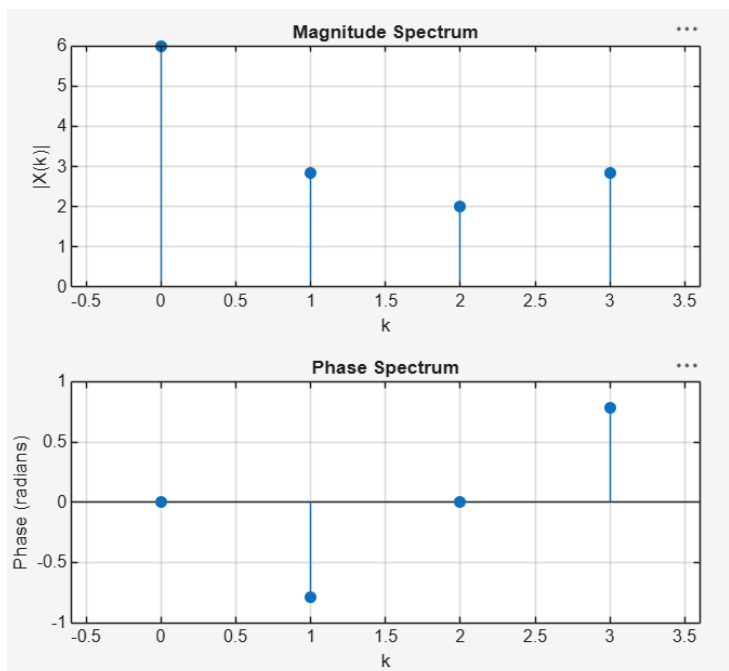
subplot(2,1,1);
stem(0:3, abs(X), 'filled');
grid on;
title('Magnitude Spectrum');
xlabel('k');
ylabel('|X(k)|');

subplot(2,1,2);
stem(0:3, angle(X), 'filled');
grid on;
title('Phase Spectrum');
xlabel('k');
ylabel('Phase (radians)');
```



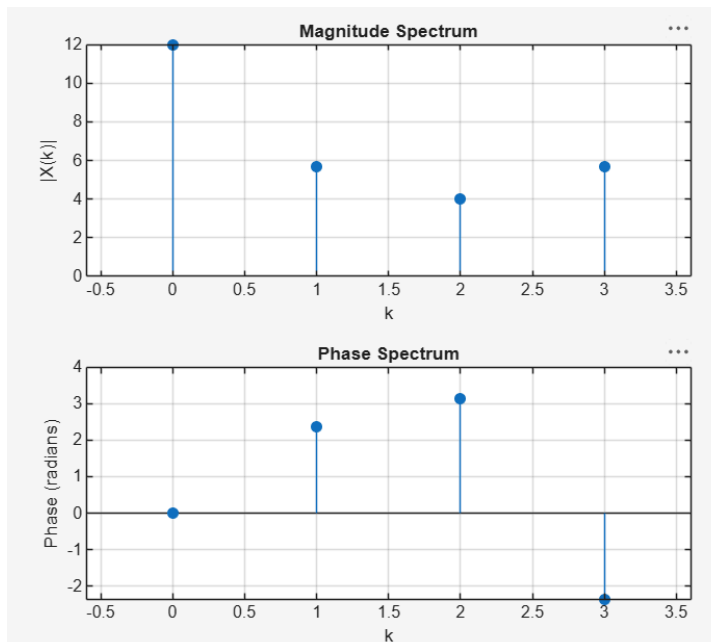
11. Find DFT of $x(n) = \{3, 2, 1, 0\}$, Plot Magnitude and Phase Spectrum of the signal.

```
1  clc;
2  clear;
3  close all;
4
5  x = [3 2 1 0];
6
7  X = fft(x,4);
8
9  disp('DFT of x(n) = [3 2 1 0]');
10 disp(X);
11
12 figure;
13
14 subplot(2,1,1);
15 stem(0:3, abs(X), 'filled');
16 grid on;
17 title('Magnitude Spectrum');
18 xlabel('k');
19 ylabel('|X(k)|');
20
21 subplot(2,1,2);
22 stem(0:3, angle(X), 'filled');
23 grid on;
24 title('Phase Spectrum');
25 xlabel('k');
26 ylabel('Phase (radians)');
```



12. Find DFT of $x(n) = \{0, 2, 4, 6\}$, Plot Magnitude and Phase Spectrum of the signal.

```
1  clc;
2  clear;
3  close all;
4
5  x = [0 2 4 6];
6
7  X = fft(x,4);
8
9  disp('DFT of x(n) = [0 2 4 6]');
10 disp(X);
11
12 figure;
13
14 subplot(2,1,1);
15 stem(0:3, abs(X), 'filled');
16 grid on;
17 title('Magnitude Spectrum');
18 xlabel('k');
19 ylabel('|X(k)|');
20
21 subplot(2,1,2);
22 stem(0:3, angle(X), 'filled');
23 grid on;
24 title('Phase Spectrum');
25 xlabel('k');
26 ylabel('Phase (radians)');
```

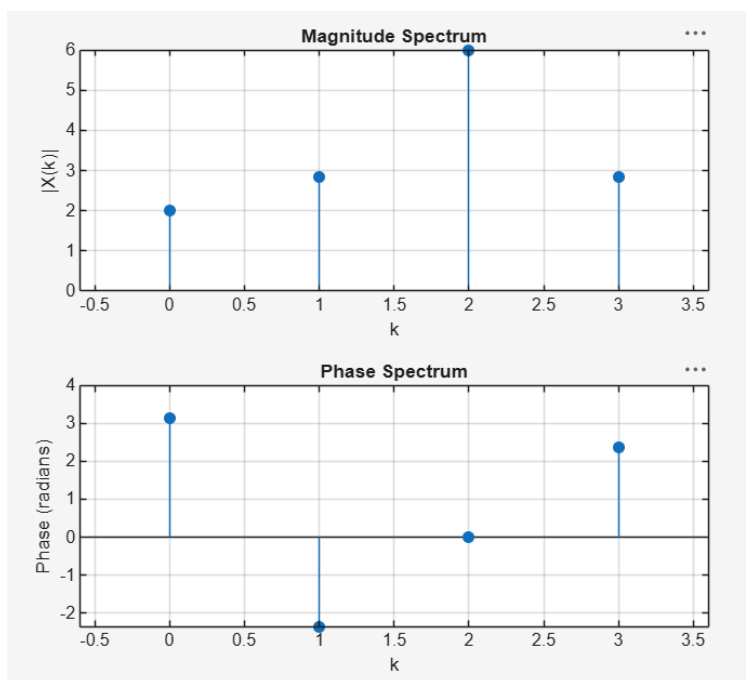


13. Find DFT of $x(n) = \{0, -1, 2, -3\}$, Plot Magnitude and Phase Spectrum of the signal.

```

1  clc;
2  clear;
3  close all;
4
5  x = [0 -1 2 -3];
6
7  X = fft(x,4);
8
9  disp('DFT of x(n) = [0 -1 2 -3]');
10 disp(X);
11
12 figure;
13
14 subplot(2,1,1);
15 stem(0:3, abs(X), 'filled');
16 grid on;
17 title('Magnitude Spectrum');
18 xlabel('k');
19 ylabel('|X(k)|');
20
21 subplot(2,1,2);
22 stem(0:3, angle(X), 'filled');
23 grid on;
24 title('Phase Spectrum');
25 xlabel('k');
26 ylabel('Phase (radians)');

```

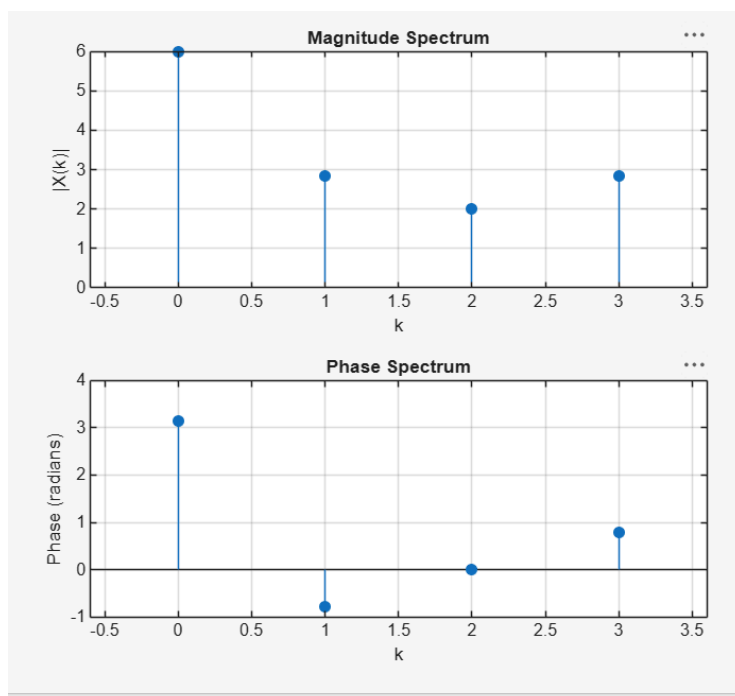


14. Find DFT of $x(n) = \{0, -1, -2, -3\}$, Plot Magnitude and Phase Spectrum of the signal.

```

1  clc;
2  clear;
3  close all;
4
5  x = [0 -1 -2 -3];
6
7  X = fft(x,4);
8
9  disp('DFT of x(n) = [0 -1 -2 -3]');
10 disp(X);
11
12 figure;
13
14 subplot(2,1,1);
15 stem(0:3, abs(X), 'filled');
16 grid on;
17 title('Magnitude Spectrum');
18 xlabel('k');
19 ylabel('|X(k)|');
20
21 subplot(2,1,2);
22 stem(0:3, angle(X), 'filled');
23 grid on;
24 title('Phase Spectrum');
25 xlabel('k');
26 ylabel('Phase (radians)');

```



15. Comment on the DFTs of 10,11,12,13,14 and 15.

- **Q10 DFT:** The sequence [3 0 1 2] is a circularly shifted version of the original sequence. The **magnitude spectrum changes slightly**, while the **phase spectrum reflects the shift** introduced in the time domain.
- **Q11 DFT:** The sequence [3 2 1 0] is the **time-reversed** version of [0 1 2 3]. Time reversal causes the **magnitude spectrum to remain the same**, while the **phase spectrum is reversed**.
- **Q12 DFT:** The sequence [0 2 4 6] is simply the original sequence **scaled by a factor of 2**. The **DFT magnitudes are doubled**, while the **phase spectrum remains unchanged**.
- **Q13 DFT:** The sequence contains **both positive and negative values**. The presence of negative samples changes both the **magnitude and phase spectra**, producing a different frequency-domain representation.
- **Q14 DFT:** The sequence [0 -1 -2 -3] is the **negative** of the original sequence [0 1 2 3]. Negating the sequence **reverses the sign of all DFT coefficients**, leaving the **magnitude spectrum unchanged** while shifting the **phase by (radians)**.

Analytical analysis and Observation: Find the DFT of the $x(n) = \{0, 1, 2, 3\}$ and using $X(k)$ thus obtained and not otherwise find the DFT of the other sequences mentioned above.

Conclusion:-

In this experiment, I learnt about how the Discrete Fourier Transform (DFT) of different discrete-time signals was successfully obtained using MATLAB. The magnitude and phase spectra were analysed for different signal operations such as zero padding, time shifting, scaling, repetition, time reversal and sign inversion. The results verified the properties of the DFT and showed how these operations affect the frequency-domain representation of the signal. Hence, the aim of analysing the frequency components of discrete-time signals using DFT was achieved successfully.

Signature of faculty in-charge